A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-75949255})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary ideal I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$I'^{\,(1-\sigma_x)^2}\,(t)\,J\mathcal{O}_{L_x} = \mathcal{O}_{L_x}, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 256 & 21 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,256 + \mathbf{Z}\,(21 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K. The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 18987314 = 0$, of discriminant -75949255. Class group invariants $(40 \quad 20 \quad 5)$, class number 4000; the three stored generator ideals (HNF) are $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 22 & 13 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 4114 & 607 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 1426)\, y^3 + (27s - 107905)\, y^2 + (-238s + 1392475)\, y + (1088s - 4891799)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 2852*y^8 - 218634*y^7 + 24019859*y^6 - 1345480831*y^5
+ 38499678307*y^4 - 599733073788*y^3 + 5185164965896*y^2 - 23454038
436920*y + 46400494202705
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{477889}{549755813888} + (-\frac{99}{1099511627776})\, s, \qquad J = \begin{pmatrix} 256 & 21 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3230601672625 = 5^3 \cdot 25844813381$:

$$\begin{pmatrix}
129224066905 & 69690603515 & 45718187850 & 106863314831 & 20146063844 & 83273874827 & 106163847728 & 114789602833 & 112945233567 & 13070778232 \\
0 & 5 & 0 & 0 & 1 & 0 & 2 & 0 & 0 & 3 \\
0 & 0 & 5 & 4 & 2 & 3 & 0 & 2 & 0 & 3 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 502 factors with signed exponents of absolute value up to 175631321380051507495382817 4; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 1426)\, y^3 + (27s - 107905)\, y^2 + (-238s + 1392475)\, y + (1088s - 4891799)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 2852*y^8 - 218634*y^7 + 24019859*y^6 - 1345480831*y^5
+ 38499678307*y^4 - 599733073788*y^3 + 5185164965896*y^2 - 23454038
436920*y + 46400494202705
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{385737}{7878081286000} + \left(\frac{43}{15756162572000}\right) s, \qquad J = \begin{pmatrix} 830 & 47 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 830$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1570108656483 = 3^2 \cdot 174456517387$:

$$\begin{pmatrix}
523369552161 & 517859494083 & 453125366563 & 499108715932 & 324933075922 & 508326874712 & 324935153520 & 218404431650 & 282732652252 & 13041597718 \\
0 & 3 & 2 & 0 & 1 & 0 & 2 & 0 & 0 & 2 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 503 factors with signed exponents of absolute value up to 1527548714546768827385452862; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 1 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 1426)\, y^3 + (27s - 107905)\, y^2 + (-238s + 1392475)\, y + (1088s - 4891799)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 2852*y^8 - 218634*y^7 + 24019859*y^6 - 1345480831*y^5
+ 38499678307*y^4 - 599733073788*y^3 + 5185164965896*y^2 - 23454038
436920*y + 46400494202705
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{460074979}{589238942107232912} + (\frac{132217}{1178477884214465824})\, s, \qquad J = \begin{pmatrix} 4114 & 607 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5527782070875 = 3^2 \cdot 5^3 \cdot 4913584063$:

$$\begin{pmatrix}
73703760945 & 19872026850 & 20064345955 & 3053963745 & 53615380030 & 40059094027 & 2925080248 & 56887145479 & 33553696778 & 39956718757 \\
0 & 15 & 10 & 11 & 9 & 10 & 0 & 3 & 10 & 11 \\
0 & 0 & 5 & 2 & 3 & 3 & 3 & 3 & 3 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 503 factors with signed exponents of absolute value up to 1221548812224684959661120760; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 0 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 1386)\, y^3 + (-19s + 160910)\, y^2 + (64s + 2446992)\, y + (1476s + 9253032)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 2772*y^8 + 319030*y^7 + 25479171*y^6 + 1180986700*y^5
+ 37075210968*y^4 + 710869269664*y^3 + 7978610479856*y^2 + 48875629
773120*y + 126997565253120
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{477889}{549755813888} + (-\frac{99}{1099511627776})\, s, \qquad J = \begin{pmatrix} 256 & 21 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $59110480752640 = 2^{12} \cdot 5 \cdot 127 \cdot 22726409$:

$$\begin{pmatrix}
115450157720 & 58604017300 & 58248579064 & 96476249458 & 14822652364 & 73070689970 & 63347750813 & 44055279435 & 105709304015 & 106048415133 \\
0 & 4 & 0 & 2 & 2 & 2 & 3 & 2 & 2 & 0 \\
0 & 0 & 4 & 0 & 2 & 0 & 1 & 2 & 0 & 2 \\
0 & 0 & 0 & 4 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 186 factors with signed exponents of absolute value up to 29464150824522635; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 1386)\, y^3 + (-19s + 160910)\, y^2 + (64s + 2446992)\, y + (1476s + 9253032)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):


```
y^10 - 2*y^9 + 2772*y^8 + 319030*y^7 + 25479171*y^6 + 1180986700*y^5
+ 37075210968*y^4 + 710869269664*y^3 + 7978610479856*y^2 + 48875629
773120*y + 126997565253120
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{385737}{7878081286000} + (\frac{43}{15756162572000})\, s, \qquad J = \begin{pmatrix} 830 & 47 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 830$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $32430087103328 = 2^5 \cdot 29 \cdot 34946214551$:

$$\begin{pmatrix} 4053760887916 & 1114975850816 & 3918827920193 & 2377368419319 & 2189792053454 & 3996667697014 & 985170487046 & 2282676270954 & 369700578806 & 3688444562001 \\ 0 & 2 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 190 factors with signed exponents of absolute value up to 32988939360536606; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 1386)\, y^3 + (-19s + 160910)\, y^2 + (64s + 2446992)\, y + (1476s + 9253032)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 2772*y^8 + 319030*y^7 + 25479171*y^6 + 1180986700*y^5
+ 37075210968*y^4 + 710869269664*y^3 + 7978610479856*y^2 + 48875629
773120*y + 126997565253120
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{460074979}{589238942107232912} + (\frac{132217}{1178477884214465824})\, s, \qquad J = \begin{pmatrix} 4114 & 607 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $28813429757440 = 2^9 \cdot 5 \cdot 29 \cdot 103 \cdot 3768077$:

$$\begin{pmatrix} 450209839960 & 198012576000 & 322992154428 & 60039223482 & 158287608112 & 427185093359 & 117977687083 & 292868893808 & 315076651153 & 138173422662 \\ 0 & 4 & 0 & 2 & 3 & 1 & 3 & 0 & 1 & 3 \\ 0 & 0 & 4 & 0 & 1 & 1 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 185 factors with signed exponents of absolute value up to 22981542407621309; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 - 1019y^3 - 37334y^2 + (-288s - 2488676)\, y + (16320s - 33863400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 - 2022*y^8 - 66516*y^7 - 3640607*y^6 + 28286772*y^5 + 6
736884636*y^4 + 254832190880*y^3 + 10297011804400*y^2 - 996762835680
0*y + 6203303989185600
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad -1 \quad 0 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{477889}{549755813888} + (-\frac{99}{1099511627776})\, s, \qquad J = \begin{pmatrix} 256 & 21 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $389865520000 = 2^7 \cdot 5^4 \cdot 11 \cdot 43 \cdot 10303$:

$$\begin{pmatrix} 97466380 & 7464400 & 15051180 & 74877597 & 83897096 & 71113476 & 29447433 & 71411485 & 93973096 & 50071820 \\ 0 & 20 & 0 & 3 & 2 & 18 & 1 & 5 & 1 & 7 \\ 0 & 0 & 10 & 4 & 4 & 8 & 7 & 0 & 7 & 4 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 5 & 3 & 4 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 126 factors with signed exponents of absolute value up to 3662; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 - 1019y^3 - 37334y^2 + (-288s - 2488676)\,y + (16320s - 33863400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 - 2022*y^8 - 66516*y^7 - 3640607*y^6 + 28286772*y^5 + 6
736884636*y^4 + 254832190880*y^3 + 10297011804400*y^2 - 996762835680
0*y + 6203303989185600
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad -1 \quad 0 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{385737}{7878081286000} + \left(\frac{43}{15756162572000}\right) s, \qquad J = \begin{pmatrix} 830 & 47 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 830$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $6956054458560 = 2^6 \cdot 3^2 \cdot 5 \cdot 2415296687$:

$$\begin{pmatrix} 289835602440 & 172671645387 & 101843854436 & 106943730740 & 92951153616 & 42442331118 & 74702172028 & 224614252324 & 31249907107 & 28319714887 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 8 & 5 & 5 & 5 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 157 factors with signed exponents of absolute value up to 1377007445; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 - 1019y^3 - 37334y^2 + (-288s - 2488676)\, y + (16320s - 33863400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 - 2022*y^8 - 66516*y^7 - 3640607*y^6 + 28286772*y^5 + 6
736884636*y^4 + 254832190880*y^3 + 10297011804400*y^2 - 996762835680
0*y + 6203303989185600
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad -1 \quad 0 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{460074979}{589238942107232912} + \left(\frac{132217}{1178477884214465824}\right) s, \qquad J = \begin{pmatrix} 4114 & 607 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $41878023644160 = 2^{10} \cdot 3^2 \cdot 5 \cdot 11^2 \cdot 1487 \cdot 5051$:

$$\begin{pmatrix} 109057353240 & 46989926744 & 92503465724 & 7378677500 & 99132730864 & 45841353691 & 35433155714 & 8179390664 & 11226486000 & 31630729491 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 4 & 0 & 0 & 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 4 & 0 & 0 & 0 & 3 & 3 & 2 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 126 factors with signed exponents of absolute value up to 14954119; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 1391)\, y^3 + (-10s - 53867)\, y^2 + (17s + 497178)\, y + (-1716s + 4919343)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 2782*y^8 - 110525*y^7 + 22022921*y^6 + 238770840*y^5
+ 5528193778*y^4 + 18818218644*y^3 + 374389109516*y^2 + 378301285147
5*y + 80102602053045
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{477889}{549755813888} + \left(-\frac{99}{1099511627776}\right) s, \qquad J = \begin{pmatrix} 256 & 21 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $582174874454229 = 3^2 \cdot 7^2 \cdot 43 \cdot 61^2 \cdot 139 \cdot 59357$:

$$\begin{pmatrix}
454469066709 & 418325050560 & 144968472069 & 104469683806 & 132469198262 & 221925824869 & 148946619689 & 14994517432 & 166511600926 & 131385562853 \\
0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\
0 & 0 & 61 & 18 & 0 & 0 & 6 & 28 & 12 & 30 \\
0 & 0 & 0 & 7 & 0 & 0 & 1 & 2 & 6 & 5 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 338 factors with signed exponents of absolute value up to 8812814898; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 1 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 1391)\, y^3 + (-10s - 53867)\, y^2 + (17s + 497178)\, y + (-1716s + 4919343)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 2782*y^8 - 110525*y^7 + 22022921*y^6 + 238770840*y^5
+ 5528193778*y^4 + 18818218644*y^3 + 374389109516*y^2 + 378301285147
5*y + 80102602053045
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{385737}{7878081286000} + \left(\frac{43}{15756162572000}\right) s, \qquad J = \begin{pmatrix} 830 & 47 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 830$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $700240148927 = 11 \cdot 63658195357$:

$$\begin{pmatrix} 700240148927 & 338912989232 & 274565176170 & 286232192234 & 235259970704 & 435050337186 & 590485335340 & 318574028349 & 23207445936 & 597468153979 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 336 factors with signed exponents of absolute value up to 7309133151; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 3 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a+b$, column e_3

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s+1391)\,y^3 + (-10s-53867)\,y^2 + (17s+497178)\,y + (-1716s+4919343)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 2782*y^8 - 110525*y^7 + 22022921*y^6 + 238770840*y^5
+ 5528193778*y^4 + 18818218644*y^3 + 374389109516*y^2 + 378301285147
5*y + 80102602053045
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{460074979}{589238942107232912} + \Big(\frac{132217}{1178477884214465824}\Big)\, s, \qquad J = \begin{pmatrix} 4114 & 607 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4114$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $12400947627875 = 5^3 \cdot 137941 \cdot 719203$:

$$\begin{pmatrix}
496037905115 & 2226323635 & 227534672730 & 342021681910 & 157748075547 & 422233141292 & 222923645571 & 388960118755 & 435359630678 & 478127784280 \\
0 & 5 & 0 & 4 & 4 & 1 & 2 & 2 & 0 & 2 \\
0 & 0 & 5 & 4 & 0 & 0 & 2 & 2 & 3 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 350 factors with signed exponents of absolute value up to 10369723651; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 3 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a+c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3180y^3 + (26s - 13)\, y^2 + 587504y + (-1438s + 719)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6360*y^8 + 11287408*y^6 + 9098898655*y^4 - 1074634422954*y^2
+ 39262802814055
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{477889}{549755813888} + (-\frac{99}{1099511627776})\, s, \qquad J = \begin{pmatrix} 256 & 21 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $16933696425491 = 19^2 \cdot 2393 \cdot 19602067$:

$$\begin{pmatrix}
891247180289 & 317727667181 & 404559778389 & 368993959401 & 176816454819 & 768037218640 & 230573207231 & 556680766635 & 392552709757 & 775171687987 \\
0 & 19 & 0 & 0 & 12 & 0 & 6 & 6 & 18 & 14 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 655 factors with signed exponents of absolute value up to 37468427207710989; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 4 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3180y^3 + (26s - 13)\, y^2 + 587504y + (-1438s + 719)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6360*y^8 + 11287408*y^6 + 9098898655*y^4 - 1074634422954*y^2
+ 39262802814055
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{385737}{7878081286000} + (\frac{43}{15756162572000})\, s, \qquad J = \begin{pmatrix} 830 & 47 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 830$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $237822829488235 = 5 \cdot 389 \cdot 1733 \cdot 70556231$:

$$\begin{pmatrix}
237822829488235 & 71205069741772 & 98654025112227 & 229416477467993 & 149955228543615 & 127768159339434 & 30462909934714 & 13827174926858 & 147450532751207 & 127671698738658 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 655 factors with signed exponents of absolute value up to 8005810598658165598; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3180y^3 + (26s - 13)\, y^2 + 587504y + (-1438s + 719)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6360*y^8 + 11287408*y^6 + 9098898655*y^4 - 1074634422954*y^2
+ 39262802814055
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{460074979}{589238942107232912} + (\frac{132217}{1178477884214465824})\, s, \qquad J = \begin{pmatrix} 4114 & 607 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $634801694441 = 103 \cdot 6163123247$:

$$\begin{pmatrix} 634801694441 & 516838584540 & 531245804914 & 186833454937 & 321737297434 & 51682868569 & 394772994018 & 219876233523 & 150488485830 & 245194352089 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 656 factors with signed exponents of absolute value up to 4658440687261638698; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 2 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 4897y^3 + (-4s + 2)\, y^2 + 11871356y + (54096s - 27048)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 9794*y^8 + 47723321*y^6 + 116571857684*y^4 + 132711991481776*
y^2 + 55564042351043520
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad 1 \quad -1 \quad 2 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{477889}{549755813888} + (-\frac{99}{1099511627776})\, s, \qquad J = \begin{pmatrix} 256 & 21 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $17084375562240 = 2^{11} \cdot 3^2 \cdot 5 \cdot 7^2 \cdot 3783211$:

$$\begin{pmatrix} 3177897240 & 1856506876 & 1867829572 & 1040419598 & 2325445912 & 1078259096 & 2907319822 & 2091901363 & 1645593617 & 2481696554 \\ 0 & 28 & 16 & 14 & 0 & 10 & 0 & 26 & 5 & 11 \\ 0 & 0 & 12 & 0 & 0 & 4 & 0 & 2 & 2 & 7 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 212 factors with signed exponents of absolute value up to 28087026; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 4897y^3 + (-4s + 2)\,y^2 + 11871356y + (54096s - 27048)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 9794*y^8 + 47723321*y^6 + 116571857684*y^4 + 132711991481776*
y^2 + 55564042351043520
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad 1 \quad -1 \quad 2 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{385737}{7878081286000} + \left(\frac{43}{15756162572000}\right) s, \qquad J = \begin{pmatrix} 830 & 47 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 830$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $52321878016000 = 2^{18} \cdot 5^3 \cdot 1596737$:

$$\begin{pmatrix} 8175293440 & 1965789460 & 7985385540 & 1259530504 & 7786920346 & 2036639997 & 7385355151 & 6709827276 & 582576653 & 6619406215 \\ 0 & 20 & 0 & 2 & 12 & 15 & 19 & 14 & 16 & 2 \\ 0 & 0 & 20 & 6 & 6 & 8 & 2 & 4 & 18 & 9 \\ 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 293 factors with signed exponents of absolute value up to 684541021; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 1 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 4897y^3 + (-4s + 2)\, y^2 + 11871356y + (54096s - 27048)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 9794*y^8 + 47723321*y^6 + 116571857684*y^4 + 132711991481776*
y^2 + 55564042351043520
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad 1 \quad -1 \quad 2 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{460074979}{589238942107232912} + \left(\frac{132217}{1178477884214465824}\right) s, \qquad J = \begin{pmatrix} 4114 & 607 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $231874899712000 = 2^{11} \cdot 5^3 \cdot 905761327$:

$$\begin{pmatrix} 36230453080 & 2972203880 & 13035000400 & 36111800288 & 3924739390 & 4538883497 & 14745563253 & 4977424729 & 15568102553 & 18817149908 \\ 0 & 10 & 0 & 4 & 4 & 5 & 2 & 3 & 6 & 1 \\ 0 & 0 & 20 & 12 & 12 & 8 & 15 & 18 & 18 & 3 \\ 0 & 0 & 0 & 4 & 2 & 1 & 2 & 3 & 0 & 3 \\ 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 212 factors with signed exponents of absolute value up to 88791; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.